Docker Generation & Testing Manual

Production Containerization, Healthcheck Probing & Edge Deployment Manual | MASTER_SDLC_AGENTS

1. Containerization Architecture

The DrowsyGuard DMS container is built using a multi-stage production Dockerfile based on Debian python:3.9-slim. The container enforces defensive isolation by executing as a non-root appuser (UID 10001) and includes automated HTTP healthcheck probes to guarantee 99.9% automotive uptime.

2. One-Command Build & Deployment Instructions

- **Build & Launch in Background:** docker compose up --build -d
- **Inspect Running Containers:** docker ps -f name=drowsyguard_dms_app
- **Stream Real-Time Container Logs:** docker logs -f drowsyguard_dms_app
- **Gracefully Stop Application:** docker compose down

3. Healthcheck Probing & Automated Recovery

The container configuration specifies an internal health probe testing http://localhost:8501/_stcore/health every 30 seconds. If the perception engine encounters an unhandled lock, Docker marks the container unhealthy and automatically restarts the service.

4. Container Resource Allocation & Constraints

Resource Attribute	Production Limit	Reservation Baseline	Purpose
CPU Allocation	2.0 Cores	0.5 Cores	Ensures sub-15ms MediaPipe perception without CPU throttling.
RAM Allocation	2048 MB	512 MB	Accommodates video buffers and prevents Out-of-Memory crashes.
Port Mapping	8501:8501	8501 (Internal)	Exposes Streamlit automotive dashboard to host network.
User Security	appuser (UID 10001)	appgroup (GID 10001)	Defensive isolation against container breakout exploits.